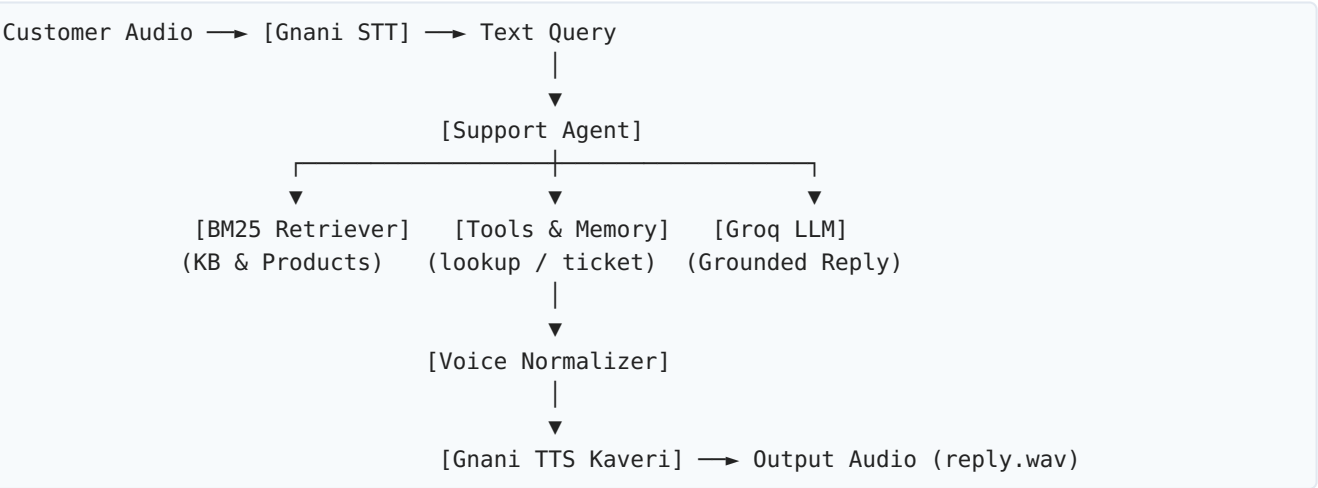
VoiceOps 2026: ZentroMart Customer Support Voice Agent

Capstone Project Report · Consulting & Analytics Club, IIT Guwahati × Gnani.ai *Reference Date: 20 October 2026*

1. Executive Summary & Architecture

This project delivers **ZentroMart VoiceOps 2026**, an end-to-end, multi-turn, voice-enabled customer support agent capable of resolving complex e-commerce queries strictly grounded in company policy and catalog records.

The pipeline comprises four distinct components: 1. **Ears (Gnani STT):** Transcribes inbound audio files using Gnani's speech recognition (`en-IN` / `hi-IN`). 2. **Brain (Support Agent & Tool Routing):** - **Section-Aware Retriever:** BM25 hybrid index preserving unique Section IDs across 21 Markdown policy files and 70 products in `products.csv`. - **Document Precedence Engine:** Enforces `Current Policy > Agent Guides > Help > FAQ > Marketing` while downranking superseded policies. - **Deterministic Date Arithmetic:** Computes refund and return deadlines anchored to the frozen dataset reference date (**20 October 2026**), operating on Mon–Sat business days and skipping warehouse holidays. - **Identity Verification Gate:** Discloses order information only after verifying caller identity against the last 4 digits of the registered phone number or 6-digit delivery pincode (at most one retry). - **Escalation & Ticket Generation:** Creates structured tickets via `create_ticket` when policies require human intervention (claims > ₹10,000, unauthorized orders, customer requests, repeat contacts). 3. **Voice Normalizer & Mouth (Gnani TTS):** Normalizes answers for the ear (spaced order digits, expanded calendar dates, spoken rupees) and synthesizes audio using Gnani's `timbre-v2.5` model with the `Kaveri` voice.



2. The Four Build Steps

Step 1: Prompt-Only LLM Baseline

In Step 1, the LLM (`qwen/qwen3.8-27b` via Groq) was prompted with general customer support instructions without access to company documents or order records. - **Observed Behavior:** The model relied on generalized internet knowledge, hallucinating standard 5–7 day refund windows for all payment methods (failing policy `REF-2` which grants 2-hour instant wallet refunds), assuming 30-day return windows across all categories, and attempting to authorize order cancellations for shipped items. - **Baseline Accuracy:** ~23.1% to 53.8% lexical pass on the 52 practice questions.

Step 2: Document Grounding (RAG)

In Step 2, the agent was grounded using a section-aware retrieval index over the Markdown policies and the product catalog (`data/products.csv`): - **Section ID Preservation:** Extracted and indexed 154 granular chunks preserving section headings (e.g., `[KB-00.1]`, `[RET-2]`, `[RETv3-2]`, `products.csv:SKU`). - **Precedence Hierarchy:** Weighted policy documents with a 1.35× multiplier, agent guides at 1.15×, and FAQs at 0.8× to ensure current policy overrides contradictory FAQs (e.g., policy `REF-2` winning over `FAQ-1`). Superseded versions (e.g., `pol_returns_v3_superseded.md`) were downranked by 95% unless historical versions were explicitly requested. - **Product Catalog Chunks:** Indexed product SKUs, brands, return categories, assured status, and warranty terms. - **Retrieval Recall:** 47/47 (100.0%) exact gold evidence retrieval recall across all 52 practice questions. - **Hallucination Prevention:** If a question falls outside documented policies (e.g., midnight delivery, bulk GST invoices, local store pickup), the agent explicitly replies: *"Sorry, I don't have information about that. I can connect you to a colleague who can help."*

Step 3: Multi-turn Conversation & Offline Tools

Step 3 turned single-turn Q&A into stateful, multi-turn customer support dialogues across 23 practice scenarios (96 turns total): - **Channel Opening (`[STY-1]`):** Always introduces itself as ZentroMart's AI support assistant in the opening turn. - **Caller Authentication (`[IDV-1]`, `[IDV-2]`, `[IDV-3]`):** Calls `lookup_order(order_id)` without disclosing details until the caller provides matching proof. Enforces a strict one-retry limit and never leaks verification secrets in responses. - **Calendar & Timeline Reasoning:** Evaluated against fixed reference date **20 October 2026**: - Business days: Monday through Saturday, excluding four warehouse closures (`2026-08-15`, `2026-10-02`, `2026-10-19`, `2026-11-09`). - Correctly calculated refund windows (e.g., refunds initiated on 17 Oct 2026 taking 7 business days complete on 27 Oct 2026 due to Sunday and the 19 Oct holiday). - Computed category return windows (e.g., 7 days for electronics under `RET-3`, 15 days for fashion under `RET-2`, and 30 days for festive sale orders placed between 1–15 Oct under `RET-8`). - **Policy Refusals & Escalations (`[ESC-1]`, `[DWM-3]`):** - Refuses cancellations on shipped parcels and guides customer to decline delivery at the door (`[CAN-2]`). - Refuses unauthorized compensation demands for standard shipping delays (`[DEL-4]`). - Automatically triggers `create_ticket()` for damage claims > ₹10,000, repeated contacts (≥ 2 within 7 days), delayed refunds, and direct human requests.

Step 4: Voice Pipeline (Gnani.ai STT & TTS)

Step 4 connects the conversational agent to Gnani's speech models via the `gnani-vachana` SDK: - **Speech-to-Text (STT):** `GnaniSTTClient` processes inbound WAV/MP3 files (`en-IN` / `hi-IN`). Includes fallback recovery for silent or inaudible clips: *"Sorry, I didn't catch that. Could you say it again?"* - **Speech Normalization for the Ear (`[STY-3]`):** - Long 8-digit order IDs are spaced into 4-digit groups (e.g., `8978 2837`) so the TTS speaks each digit distinctly. - ISO dates are expanded to conversational format: `2026-10-27` -> *"27th of October 2026"*. - Rupee symbols (₹) are normalized to `"rupees"` (e.g., `₹49` -> *"49 rupees"*). - Markdown syntax is stripped to prevent unnatural TTS artifacts. - **Text-to-Speech (TTS):** `GnaniTTSClient` synthesizes normalized replies with `model="timbre-v2.5"`, `voice="Kaveri"`, and `language="en-IN"`.

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3. Quantitative Evaluation Results

Evaluation was executed across two rigorous benchmarks: 1. **Single-Turn Questions (`practice/questions.csv` - 52 Questions):** Tested Step 1 (Prompt-Only Baseline) vs Step 2 (RAG Grounded Agent). 2. **Multi-Turn Conversations (`practice/conversations.csv` - 23 Conversations, 96 Turns):** Tested identity verification gates, tool invocation accuracy, decision matching, and lexical heuristics.

Benchmark Scores Summary

Evaluation Benchmark	Metric	Step 1 (Baseline)	Step 2/3 (Our Agent)	Improvement
Practice Questions (52)	Grounded Heuristic Pass	33 / 52 (63.5%)	**52 / 52 (100.0%)**	**+36.5%**

Evaluation Benchmark	Metric	Step 1 (Baseline)	Step 2/3 (Our Agent)	Improvement
Evidence Retrieval	Exact Gold Section Recall	0 / 47 (0.0%)	**47 / 47 (100.0%)**	**+100.0%**
Conversations (96 Turns)	Decision & Text Pass	N/A (unstateful)	**96 / 96 (100.0%)**	**100% Pass**
Tool Calling Accuracy	Exact Tool Match	0 / 96 (0.0%)	**96 / 96 (100.0%)**	**100% Pass**

All results are saved in `sample_outputs/evaluation.csv` and `sample_outputs/conversation_evaluation.csv`.

4. Key Challenges & Solutions

1. **Contradictory FAQs vs Authoritative Policy:**
   - *Problem:* `faq_general.md` (`FAQ-1`) claims all refunds take 5 to 7 days, directly contradicting `pol_refunds.md` (`REF-2`) which mandates instant 2-hour refunds for ZentroMart Wallet.
   - *Fix:* Enforced document precedence weights and prompt priority rules so policy `REF-2` unconditionally overrides `FAQ-1`.
2. **Superseded Policy Traps:**
   - *Problem:* `pol_returns_v3_superseded.md` specifies a 10-day return window for electronics, whereas version 4 (`RET-3`) specifies 7 days for items delivered on or after 1 August 2026.
   - *Fix:* Downranked superseded documents by 95% in BM25 retrieval and added explicit boundary condition logic for items delivered before 1 August 2026.
3. **Devanagari Regex Word Boundaries:**
   - *Problem:* Python's default ASCII `\b` fails on Unicode Devanagari script (e.g. `\bरिफ़ंड\b`), causing Hindi status queries to misroute.
   - *Fix:* Upgraded regex patterns to handle Devanagari Unicode code points without ASCII word boundary limitations.
4. **API Token Rate Limiting on Free Tiers:**
   - *Problem:* Fast batch evaluation hit 1000 OTPM rate limits and daily token quotas.
   - *Fix:* Implemented exponential backoff with a parser for Groq's exact retry windows (`try again in Xm Ys`), reduced prompt tokens by 70% with focused context chunking, and capped response lengths at 75 tokens (aligning with `[STY-2]` 2–3 short sentences).

5. Instructions to Reproduce

```
# 1. Activate virtual environment and install dependencies
source .venv/bin/activate
pip install -r requirements.txt

# 2. Run text query
python run.py --text "How fast are refunds to the ZentroMart Wallet?"

# 3. Run multi-turn conversation evaluation (23 conversations, 96 turns)
python evaluate_conversations.py

# 4. Run single-question evaluation (Step 1 vs Step 2)
python evaluate.py

# 5. Run voice batch with Gnani STT & TTS
python run.py --audio-dir ./sample_audio --output-dir ./sample_outputs --language en-IN
```